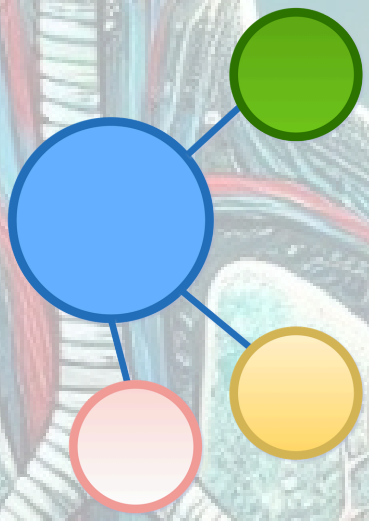
Installation guide



OrthoGather

A local platform for orthology-based proteome comparisons and Gene Ontology enrichment

Version 2.2.0 · October 2026

Carlos Vivas-Rodríguez, David Gomez Matallanas, Colm J. Ryan, Siobhán McClean, Olivier Dennler and Joanna Drabińska

github.com/CarlosVivasR/OrthoGather · doi.org/10.5281/zenodo.18510910

UCD CANCER DATA LAB

University College Dublin
Ireland's Global University

Contents

1. Before you start
2. Installer for macOS
3. Installer for Windows
4. Which version is installed
5. Conda: macOS, Linux and WSL
6. Other ways to install
7. Troubleshooting
8. Uninstalling

This guide covers installation only. How to use OrthoGather is described in the README, the tutorial inside the application and the paper: Vivas-Rodríguez C, Gomez Matallanas D, Ryan CJ, McClean S, Dennler O, Drabinska J. OrthoGather: a local platform for orthology-based proteome comparisons and Gene Ontology enrichment. bioRxiv (2026), doi.org/10.64898/2026.01.30.702851.

Questions and problems: github.com/CarlosVivasR/OrthoGather/issues

Before you start

Disk space

About 3 GB on macOS and 5 GB on Windows, where a small Linux system (WSL) is added. The first installation downloads 1–2 GB.

Memory

8 GB of RAM or more. OrthoFinder is the step that uses most memory and time.

System

macOS on Apple Silicon or Intel; Windows 10 version 2004 or later, or Windows 11, with administrator rights; Linux through Conda.

Internet connection

To install, to download proteomes and GO annotation files, and to load the plotting libraries. The analyses themselves run on your computer.

Installer for macOS

Download Install-OrthoGather-Mac.zip

1 Unzip

Double-click the downloaded `Install-OrthoGather-Mac.zip` (usually in your Downloads folder). You get a file called `Install OrthoGather.command`.

2 Open it and allow it once

Double-click `Install OrthoGather.command`. Because the installer is not signed, macOS warns that it cannot verify the file. Close the warning, open **System Settings > Privacy & Security**, scroll down to the message about `Install OrthoGather.command` and click **Open Anyway**, confirming if asked. This is needed only once. On older versions of macOS you can instead Control-click the file and choose **Open**.

3 Let it run

A Terminal window opens and shows the progress. The installer sets up Miniforge (Conda) if it is not already present, creates an environment with Python 3.11, OrthoFinder 2.5.5 and the other dependencies, and downloads the latest published version of OrthoGather from GitHub. This takes about 5–15 minutes, and the window can look idle for a minute or two at a time. Leave it open until it shows **OrthoGather is installed!** or an error message.

Along the way you may be asked three things. If Conda is not installed, the installer asks whether to install Miniforge (about 400 MB, in your home folder): press Return to accept. macOS may ask whether Terminal can access your Desktop or other folders: allow it, because the launcher is created there. At the end, the installer offers to start OrthoGather.

4 Start OrthoGather

The installer adds an **OrthoGather** app, with the OrthoGather logo, to your Desktop and to the Applications folder in your home folder. Double-click it whenever you want to use OrthoGather; it opens in your web browser, usually at `http://127.0.0.1:5000`, or at `http://127.0.0.1:5001` if AirPlay Receiver is using port 5000. Closing the browser tab does not stop OrthoGather: open the same address to return to it.

On macOS, OrthoGather runs without a window of its own. To stop it, choose **Force Quit** from the Apple menu and select OrthoGather, or restart the computer.

Installer for Windows

OrthoFinder has no native Windows build, so the Windows installer sets up the Windows Subsystem for Linux (WSL) with Ubuntu and installs OrthoGather inside it. It needs administrator rights, hardware virtualisation enabled in the BIOS, and one restart.

Download Install-OrthoGather-Windows.zip

1 Extract the zip

Right-click `Install-OrthoGather-Windows.zip`, choose **Extract All** and open the extracted folder, which contains `Install OrthoGather.bat`. Do not run the installer from inside the zip preview window, as it can fail.

2 Run it

Double-click `Install OrthoGather.bat`. If a "Windows protected your PC" message appears, click **More info > Run anyway**, then **Yes** on the permission prompt. The warning appears because the installer is not signed.

3 Restart when asked

If WSL has to be enabled, the installer asks to restart the computer; save your work and press **Y**. After the restart it should reopen by itself and ask once more for administrator permission: click **Yes** to let it continue. If it has not reopened a minute or two after you log in, run `Install OrthoGather.bat` again: it resumes where it stopped.

4 Let it run

The installer installs Ubuntu and downloads the scientific packages and OrthoGather, which takes about 5–15 minutes. There is no Linux username or password to create. Leave the window open until it shows **OrthoGather is installed!** or an error message. If the window closes unexpectedly, run the installer again.

5 Start OrthoGather

Double-click the **OrthoGather** shortcut on the Desktop. A small window opens and, a few seconds later, the app appears in your web browser.

> You install only once. On Windows, a small OrthoGather window stays open while you work: that is the application running, and closing it stops OrthoGather. To update, on either system, run the installer again; it refreshes the installation and keeps your data.

Which version is installed

The installers download the latest published version of OrthoGather from GitHub, the one shown in the latest release. To install a specific version, for example 2.1.0, the version described in the preprint, use the Conda route below and clone that release:

```
git clone --branch v2.1.0 https://github.com/CarlosVivasR/OrthoGathe
r.git
```

The version in use is shown in the application's sidebar and in every provenance file it exports. A clone of a release is not on a branch, so to move to another version later, run `git fetch --tags` and `git checkout` with its tag (for example `v2.2.0`), then `conda env update -f environment.yml --prune`.

Conda: macOS, Linux and WSL

For users who prefer the command line, and the supported route on Linux.

Conda or Micromamba

The package manager that builds the environment. If you do not have one, install **Miniforge**.

Git

To download the code. A plain clone is enough: the proteome catalogue is stored compressed in the repository, so Git LFS is not needed.

1 Download OrthoGather

```
git clone --branch v2.2.0 https://github.com/CarlosVivasR/OrthoGather.git
cd OrthoGather
```

2 Create the environment

This installs Python 3.11, OrthoFinder 2.5.5 and the Python dependencies, and runs natively on Apple Silicon. The first build takes a few minutes.

With Conda

```
conda env create -f environment.yml
conda activate orthogather
```

With Micromamba

```
micromamba env create -f environment.yml
micromamba activate orthogather
```

3 Start OrthoGather

The application picks a free port and opens your browser. To stop it, press **Ctrl+C** in the terminal.

```
python app.py
```

On the first launch, the proteome catalogue (about 260 MB) is unpacked from its compressed copy. This happens once and takes a few seconds. OrthoGather also checks GitHub for a newer catalogue and offers to update it.

OrthoGather opens in your browser. From there, choose **New Analysis**, the **Preselected Dataset** or **External Upload**.

To start it again later, from the `OrthoGather` folder:

```
conda activate orthogather
python app.py
```

To update to a newer release, from the `OrthoGather` folder, with its tag in place of `v2.2.0`:

```
git fetch --tags
git checkout v2.2.0
conda env update -f environment.yml --prune
```

Other ways to install

Windows without the installer

Install **WSL** first, then follow the Conda steps inside it.

pip

With Python 3.11, `pip install -r requirements.txt` installs the Python packages, but OrthoFinder 2.5.5 is not on PyPI and must be installed separately. Conda is the supported route.

Troubleshooting

macOS will not open the installer

Open **System Settings > Privacy & Security**, scroll down to the message about the file and click **Open Anyway**. On older versions of macOS, Control-click the file and choose **Open**.

If both are blocked, open Terminal and run the line below, changing the path if the file is not in Downloads:

```
xattr -dr com.apple.quarantine ~/Downloads/"Install OrthoGathe
r.command"
chmod +x ~/Downloads/"Install OrthoGather.command"
```

Windows shows "Windows protected your PC"

Click **More info**, then **Run anyway**. The message appears because the installer is not code-signed.

The antivirus blocked or deleted the download

Some managed or work computers block downloads of program files, which is why the installers are distributed as .zip files. If the download is still blocked, or the file disappears after downloading, it is your organisation's security policy: ask your IT support to allow it, or open an issue on GitHub.

Windows says Linux cannot start, or virtualisation is off

WSL needs hardware virtualisation switched on in the computer's BIOS/UEFI, and this is the one step an installer cannot do for you. Restart the computer, open the BIOS (usually by pressing F2, F10, Del or Esc as it starts), find the setting called Virtualization Technology, Intel VT-x, AMD-V or SVM, set it to Enabled, save and exit, then run the installer again.

The installer keeps a log at `%LOCALAPPDATA%\OrthoGather\install-log.txt`; attach it if you report a problem.

The installer seems frozen

The first installation downloads large scientific packages and can stay quiet for a minute or two at a time. Wait until it shows **OrthoGather is installed!** or an error message. Closing the window is the only thing that cancels it.

The browser did not open

Open `http://localhost:5000` in your browser. If port 5000 was busy, OrthoGather uses the next free port (5001, 5002 and so on). On Windows, the small OrthoGather window shows the address; keep that window open while you use OrthoGather.

macOS: OrthoGather opens on port 5001

AirPlay Receiver, which is on by default in recent versions of macOS, uses port 5000, so OrthoGather runs on the next free port, usually `http://127.0.0.1:5001`. This does not affect the analyses. To keep port 5000 for OrthoGather, turn off AirPlay Receiver in **System Settings > General > AirDrop & Handoff**.

`conda: command not found`

Conda is not installed, or your terminal has not loaded it. Install Miniforge, close and reopen the terminal, and try again.

The species list is empty or the catalogue did not load

The catalogue is unpacked from its compressed copy on first launch, so this is rare. If it happens, stop OrthoGather, delete the half-written file with the command for your installation, and start OrthoGather again as usual; it unpacks the catalogue again from `proteomes_list.json.gz`.

Conda installation (from the OrthoGather folder):

```
rm -f static/Proteomes_json/proteomes_list.json
```

macOS installer (in Terminal):

```
rm -f ~/Applications/OrthoGather/static/Proteomes_json/proteome
s_list.json
```

Windows installer (in Command Prompt):

```
wsl -d Ubuntu -u root -- rm -f /root/OrthoGather/static/Proteom
es_json/proteomes_list.json
```

If that does not help, open the catalogue menu on the species page and click **Check for updates** to download a fresh copy from GitHub.

`ModuleNotFoundError` or "OrthoFinder not found" on start

The environment is not active, or it did not finish building. Run `conda activate orthogather` (the prompt should then show `(orthogather)`); if the error remains, run `conda env update -f environment.yml --prune` from the OrthoGather folder and start OrthoGather again.

OrthoFinder is slow or uses too much CPU

OrthoGather uses all logical CPUs for the sequence search. To leave some free for other work, start it from the OrthoGather folder of a Conda installation with lower thread counts:

```
ORTHOGATHER_OF_THREADS=6 ORTHOGATHER_OF_ANALYSIS_THREADS=1 pyth
on app.py
```

Uninstalling

macOS installer

Delete `OrthoGather.app` from the Desktop and from the Applications folder in your home folder, the folder `~/Applications/OrthoGather`, and the environment with `conda env remove -n orthogather`. If Miniforge was installed only for OrthoGather, run `conda init --reverse bash zsh` and then delete `~/miniforge3`.

Windows installer

Delete the OrthoGather shortcut on the Desktop and the folder `%LOCALAPPDATA%\OrthoGather`. If Ubuntu was installed only for OrthoGather, this removes it completely, including everything stored in it:

```
wsl --unregister Ubuntu
```

If you keep Ubuntu, remove OrthoGather from it instead, and `/root/miniforge3` as well if nothing else uses it:

```
wsl -d Ubuntu -u root --
rm -rf /root/OrthoGather
/root/.orthogather
```

Conda installation

Delete the cloned `OrthoGather` folder and remove the environment with `conda env remove -n orthogather`.

Saved analyses

The analysis history is kept in `~/.orthogather` (inside Ubuntu on Windows). Delete it only if you no longer need your saved runs.